BUILD A BIN THAT THINKS.

SMART TRASH BIN

Arduino • Sensors • Servo Control • Product Design

MASKIDO INNOVATION ACADEMY

PROJECT 01

Instructor Delivery Guide

Preparation, facilitation, troubleshooting and assessment.

INSTRUCTOR EDITION

Designed for small groups, school clubs and subscriber-supported home learning.

Delivery at a Glance

Recommended setup	Guidance
Learner age	9–14; adjust cutting responsibility by age and policy
Group size	2–4 learners per kit
Total time	4–6 hours; recommended as four 60–90 minute sessions
Required supervision	Adult/instructor for cutting, hot glue, power inspection and first switch-on
Success standard	10 reliable open-close cycles + learner explanation + evidence upload

Non-negotiable technical standard
Use a standardised, protected 5V supply rated at 2A minimum for the servo. Do not improvise with loose lithium cells. Do not use an L298N in this build.

Before class

- Build and test one complete reference prototype using the exact kit.
- Confirm the supported Arduino IDE/board setup and test the .ino file.
- Label all kit bags and colour-code wires.
- Pre-cut panels where learners are not permitted to use cutting tools.
- Charge/prepare approved power sources.
- Print workbooks or provide editable digital copies.
- Prepare a safe “power inspection” station.

Recommended Session Plan

Session	Time	Missions	Main outcome
1 · Discover	60–75 min	01–03	Problem, component knowledge and safety gate
2 · Experiment	75–90 min	04–05	Sensor readings and servo calibration
3 · Build	90–120 min	06–07	Working bench circuit and enclosure
4 · Finish	75–90 min	08	Installation, 10-cycle test, quiz and evidence

Facilitation method

- Ask before telling: let learners predict the purpose of each component.
- Test one subsystem at a time: sensor → servo → combined circuit → mechanical build.
- Make evidence visible: every mission ends with a recorded result.
- Praise the debugging process, not only the final working product.
- Use correct terms after the analogy: sensor, microcontroller, actuator, regulated supply and shared ground.

Language balance
"Eyes, brain and muscle" should open the idea. Accurate engineering terminology should complete it.

Kit and Preparation Checklist

Item	Per team	Instructor check
Arduino Uno-compatible board + USB cable	1	◻
HC-SR04 ultrasonic sensor	1	◻
SG90 micro servo	1	◻
Approved regulated 5V supply, 2A minimum	1	◻
Breadboard + jumper wires	1 set	◻
5 mm foam board / pre-cut panels	1 set	◻
Light hinge tape + linkage material	1 set	◻
Removable mounting tape / cable ties	As needed	◻
Ruler, pencil and cutting mat	Shared	◻
Craft knife and hot-glue gun	Adult station only	◻

Wiring approval checklist

- Sensor VCC/GND correct.
- TRIG on D10 and ECHO on D9.
- Servo signal on D11.
- Servo powered from approved external 5V.
- Arduino GND and external supply GND connected.
- No exposed conductors touching.
- Mechanical linkage disconnected for first powered test.

Common Failure Guide

Symptom	Diagnosis sequence	Do not do
No sensor reading	Check VCC/GND → D10/D9 → Serial baud → sensor direction	Do not replace everything at once
Servo jitters or resets board	Check external 5V rating → shared GND → tight wires → mechanical load	Do not increase voltage
Servo hums at closed position	Reduce closed angle or reposition horn with power off	Do not let it force a hard stop
False opening	Move sensor away from lid/linkage → tighten wires → test environment	Do not hide the problem with a long delay
Lid too heavy	Reduce lid mass/friction or use suitable servo after design review	Do not overload an SG90

Debugging rule
Change one variable at a time and record what changed. This turns troubleshooting into learning.

Mechanical inspection

- Lid rotates freely by hand with power disconnected.
- Servo horn does not touch foam at any angle.
- Linkage has slight play and does not bind.
- Electronics stay clear of waste and can be removed.
- Wires have strain relief and do not cross the moving arm.

Assessment Rubric · 20 Marks

Criterion	4	3	2	1
Function	10 reliable cycles	Mostly reliable	Works with support	Not functional
Safety + wiring	Accurate, neat, safe	Minor clutter	Corrected errors	Unsafe/incomplete
Code understanding	Explains thresholds, hold, servo	Explains main logic	Names some parts	Cannot explain yet
Product finish	Clean, sturdy, serviceable	Good overall	Rough but usable	Fragile/inaccessible
Reflection	Clear problem + improvement	Explains solved problem	Names challenge	No reflection

Evidence standard

- One well-lit photo of the finished prototype.
- One continuous 30-second demo showing approach, opening, use, retreat and closing.
- Learner explains one fault discovered and how it was corrected.
- Instructor verifies safety and function before badge approval.

Badge decision

Award the Smart Systems Builder badge at 14/20 or higher, provided safety and functional evidence are both satisfactory.

Content Production Notes

Asset	Production direction
Project trailer	Actual Maskido prototype and real learner hands; 20–30 seconds; energetic, clean and truthful
Component visuals	One consistent photographic or 3D style on neutral branded backgrounds
Wiring video	Overhead camera, clear FROM → TO labels, no jump cuts during critical connections
Construction video	Show dimensions, adult cutting, square assembly, access panel and finishing
Captions	Open captions for key steps plus accurate transcript
Thumbnail	Finished prototype, learner reaction, large “SMART BIN” title, minimal clutter

Copyright and trust
Use original Maskido footage and licensed assets only. The product shown at the beginning must match the prototype the learner will actually build.

Instructor closing script

“You did more than assemble components. You identified a problem, built a system, tested it repeatedly and improved it. That is what engineers do. Your next mission is to make this invention even more useful.”